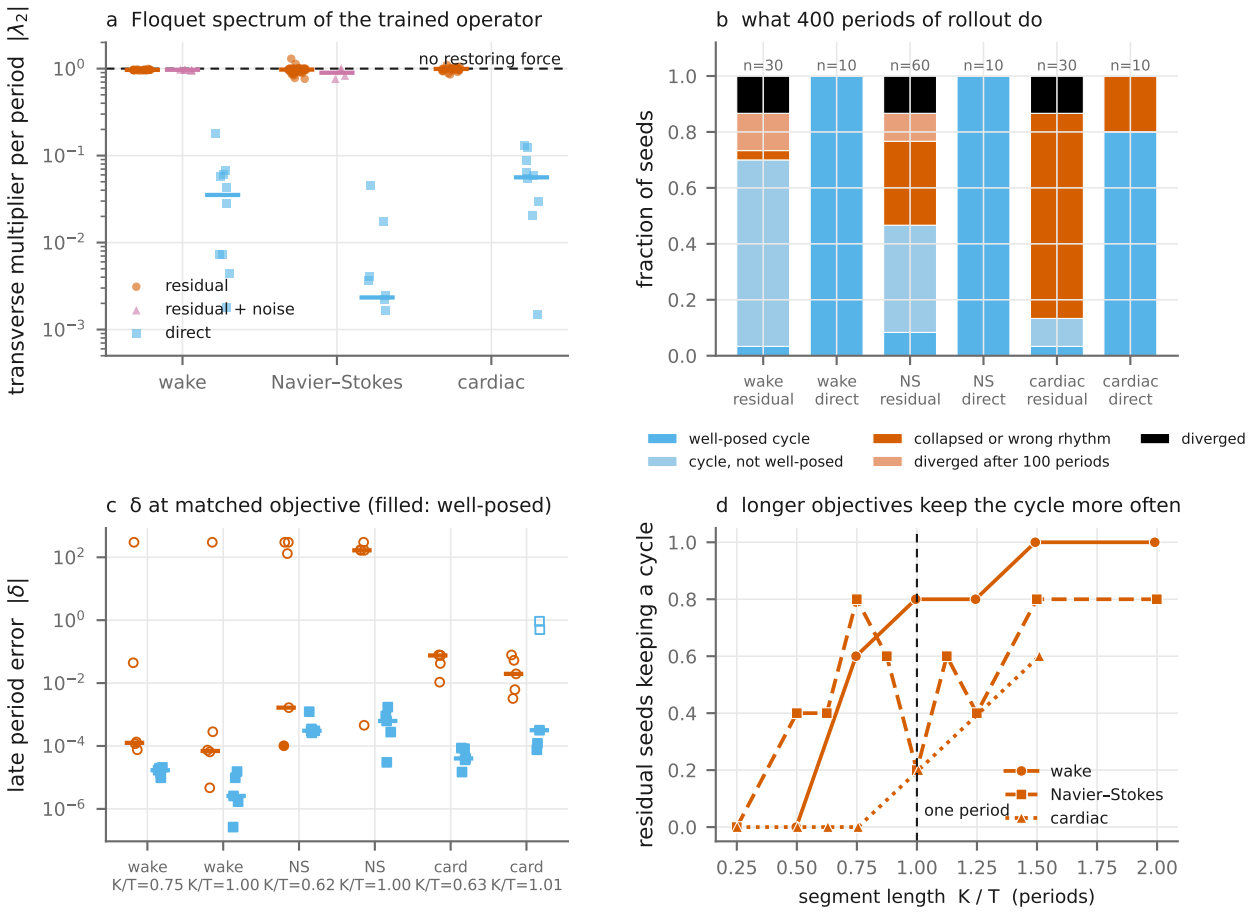
a Floquet spectrum of the trained operator
transverse multiplier per period |λ2|
no restoring force
residual
residual + noise
direct
wake
Navier–Stokes
cardiac
b what 400 periods of rollout do
fraction of seeds
n=30
n=10
n=60
n=10
n=30
n=10
wake residual
wake direct
NS residual
NS direct
cardiac residual
cardiac direct
well-posed cycle
cycle, not well-posed
collapsed or wrong rhythm
diverged after 100 periods
diverged
c δ at matched objective (filled: well-posed)
late period error |δ|
wake K/T=0.75
wake K/T=1.00
NS K/T=0.62
NS K/T=1.00
card K/T=0.63
card K/T=1.01
d longer objectives keep the cycle more often
residual seeds keeping a cycle
one period
wake
Navier–Stokes
cardiac
segment length K / T (periods)